$$\varphi(r_2) = \frac{Q}{4\pi\varepsilon_0}\left(\frac{1}{r_2} - \frac{1}{\infty}\right).$$

$$\varphi(r_2) = \frac{Q}{4\pi\varepsilon_0}\cdot\frac{1}{r_2}$$

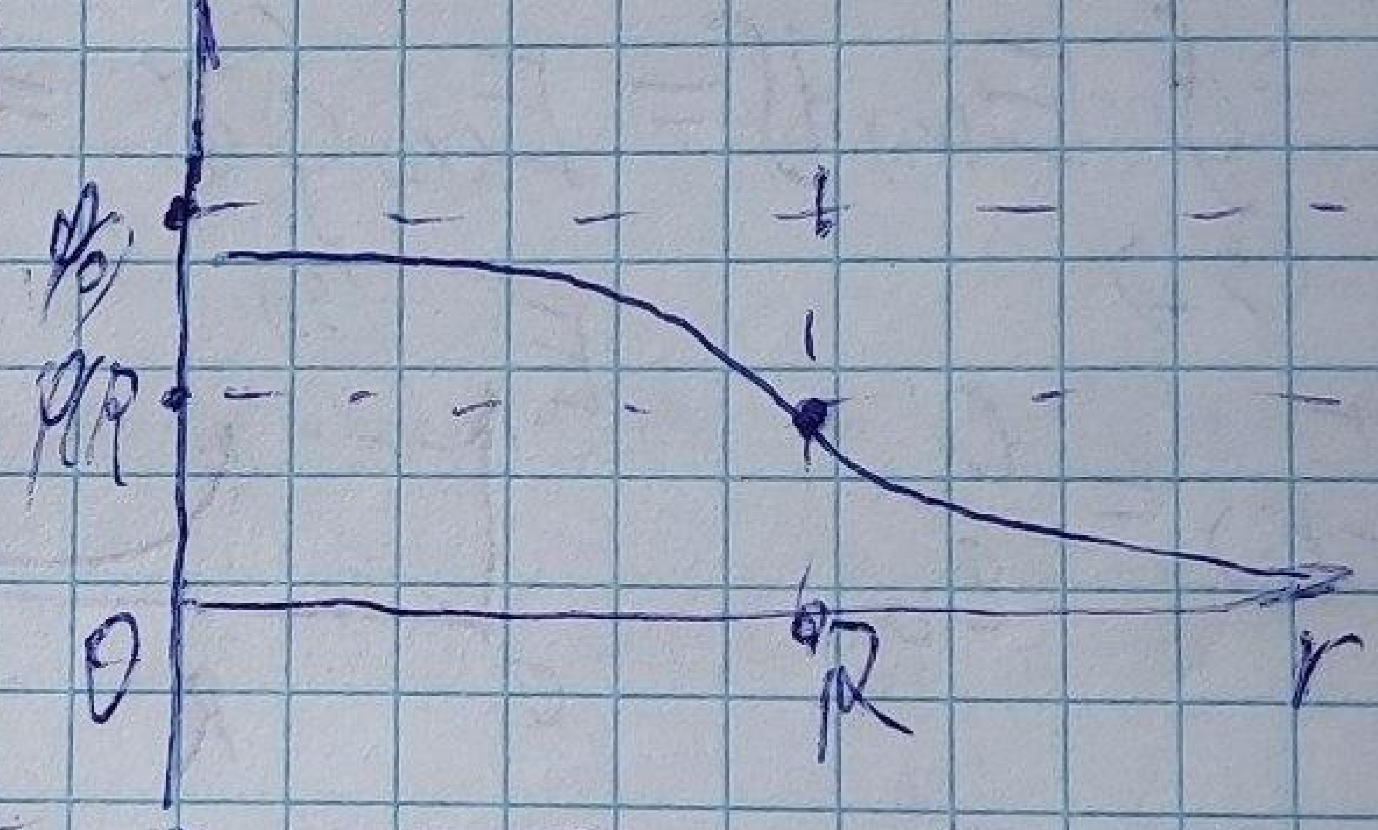
$$\int_{\varphi(R)}^{\varphi(r_1)} d\varphi = -\int_R^{r_1} \frac{Qr}{4\pi R^3\varepsilon_0}\, dr;$$

$$\varphi(r_1) - \varphi(R) = -\frac{Q}{4\pi R^3\varepsilon_0}\left(\frac{r^2}{2}\right)\Big|_R^{r_1}$$

$$\varphi(r_1) - \varphi(R) = -\frac{Q}{4\pi R^3\varepsilon_0}\left(\frac{r_1^2 - R^2}{2}\right), \quad \varphi R = \frac{Q}{4\pi\varepsilon_0 R}$$

$$\varphi(r_1) = \frac{Q(3R^2 - r_1^2)}{8\pi R^3\varepsilon_0}$$

$$\varphi(0) = \frac{3Q}{8\pi R\varepsilon_0}$$



$\vec{j} = qn\vec{v}$; $\vec{j} = \sigma\vec{E}$ – закон Ома в локальной ф

$\vec{E} = -\nabla\varphi$

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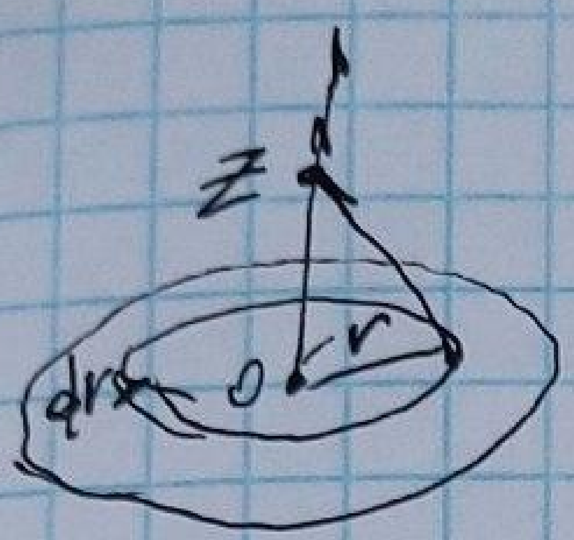
$$\vec{j} = \sigma\vec{E} = en\vec{v}$$

П.з.



а) $d\varphi = \frac{k\,dq}{\sqrt{r^2+z^2}}$; $\sigma = \frac{dq}{dS}$; $dq = \sigma\,dS$

$\sigma = \frac{Q}{\pi R^2}$; $dS = 2\pi r\,dr$;

$$d\varphi = \frac{k\sigma\,dS}{\sqrt{r^2+z^2}} = \frac{kQ\cdot 2\pi r\,dr}{\pi R^2\sqrt{r^2+z^2}};$$

$$\int_0^R d\varphi = \int_0^R \frac{kQ\cdot 2r\,dr}{R^2\sqrt{r^2+z^2}};$$

$$\varphi = \frac{2kQ}{R^2}\int \frac{r\,dr}{\sqrt{r^2+z^2}} = \left| \begin{matrix} u = r^2+z^2 \\ du = 2r\,dr \end{matrix} \right. = \frac{2kQ}{R^2}\int_{z^2}^{R^2+z^2} \frac{u^{-\frac{1}{2}}\,du}{2} =$$

$$= \frac{kQ}{R^2}\cdot u^{\frac{1}{2}}\cdot 2\cdot \Big|_{z^2}^{R^2+z^2} = \frac{2kQ}{R^2}\left(\sqrt{R^2+z^2}-\sqrt{z^2}\right)$$

б) $\vec{E} = -\vec{\nabla}\varphi = -\bar{k}\,\frac{\partial\varphi(z)}{\partial z}$;

$$\bar{E} = -\bar{k}\,\frac{\partial\varphi}{\partial z} = -\bar{k}\left(\frac{2kQ}{R^2}\left(\sqrt{R^2+z^2}-\sqrt{z^2}\right)\right)' =$$

$$= -\bar{k}\cdot\frac{2kQ}{R^2}\left(\frac{z}{\sqrt{R^2+z^2}}-1\right)$$